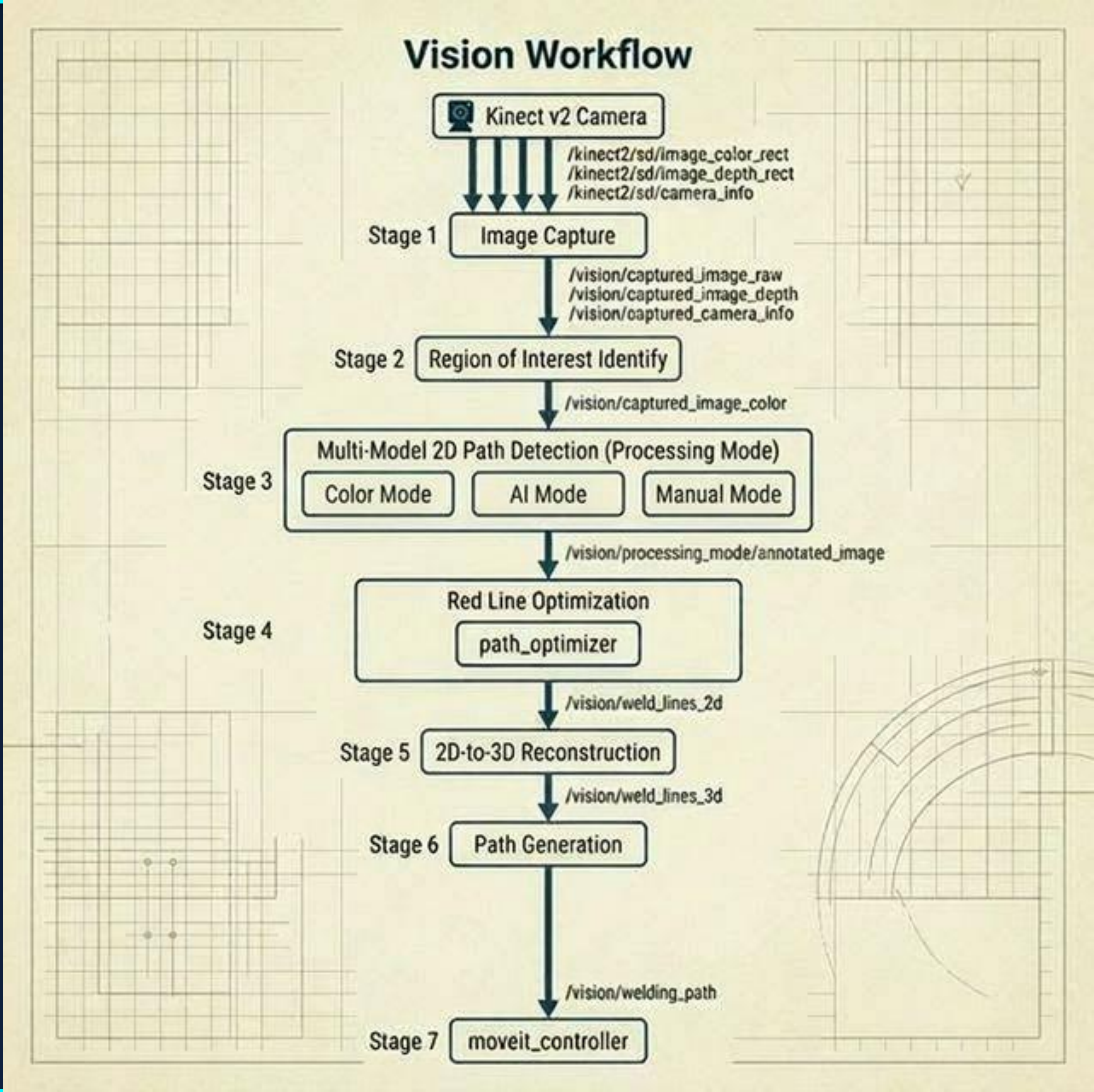
Vision System

Vision Workflow Diagram:

The ASPIRE pipeline is organized as an **asynchronous ROS 2 DAG**: distinct nodes execute concurrently and communicate over ROS topics. There are no cycles, so each node simply subscribes to input topics and publishes to output topics.



Stage 1: Image Capture

The color image

The depth image

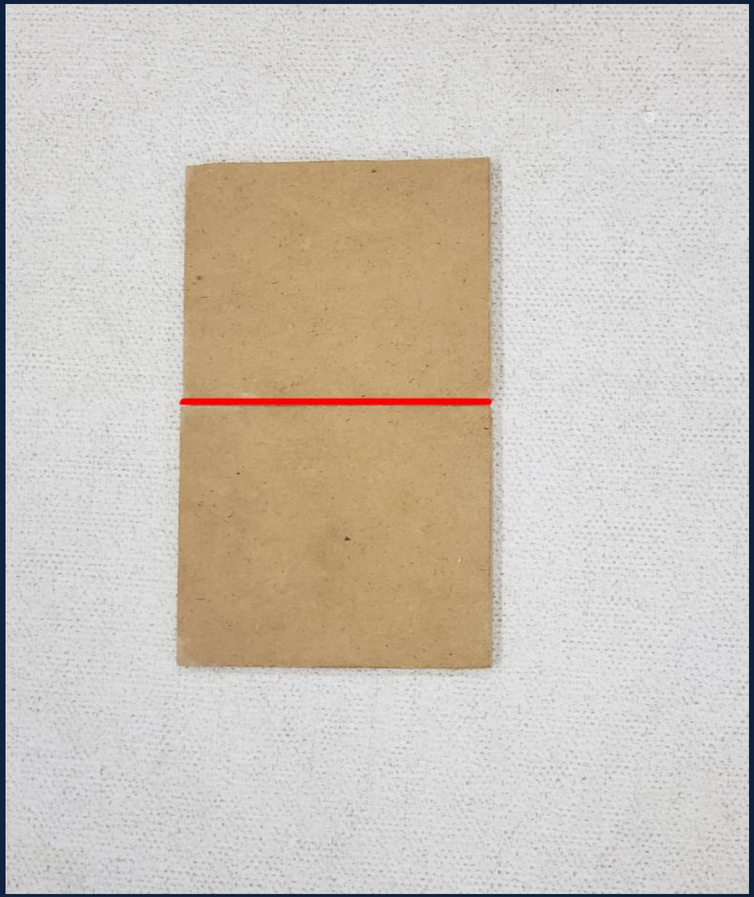
Stage 2: Region of Interest Identify

The captured raw image

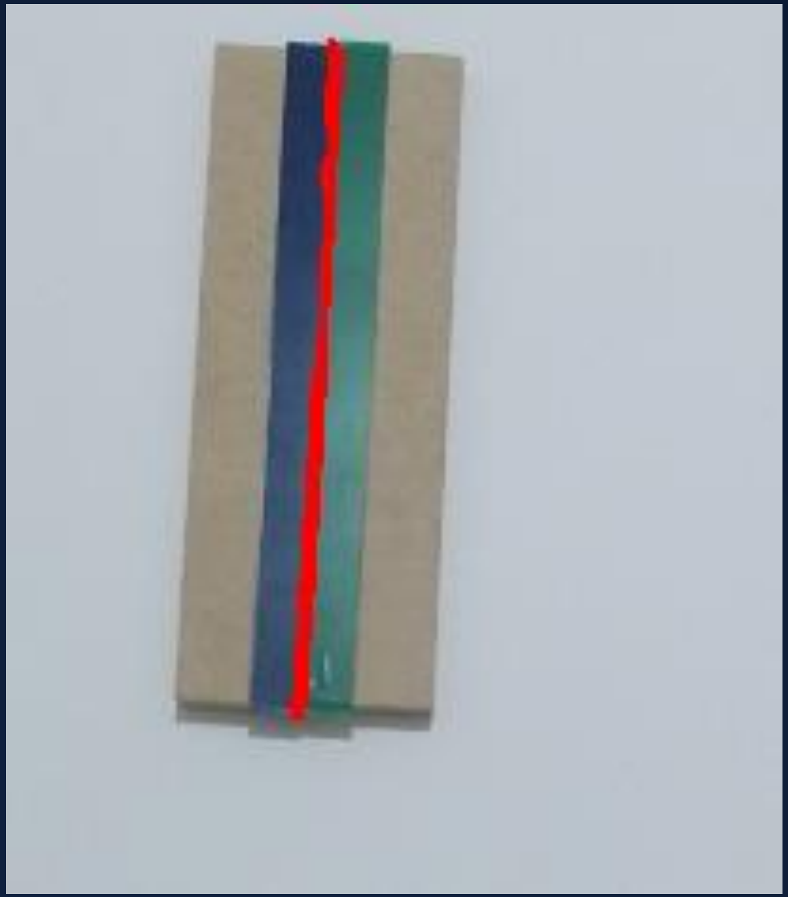
The isolated workspace

Stage 3: Multi-Model 2D Path Detection (Processing Mode)

AI Mode



Color Mode

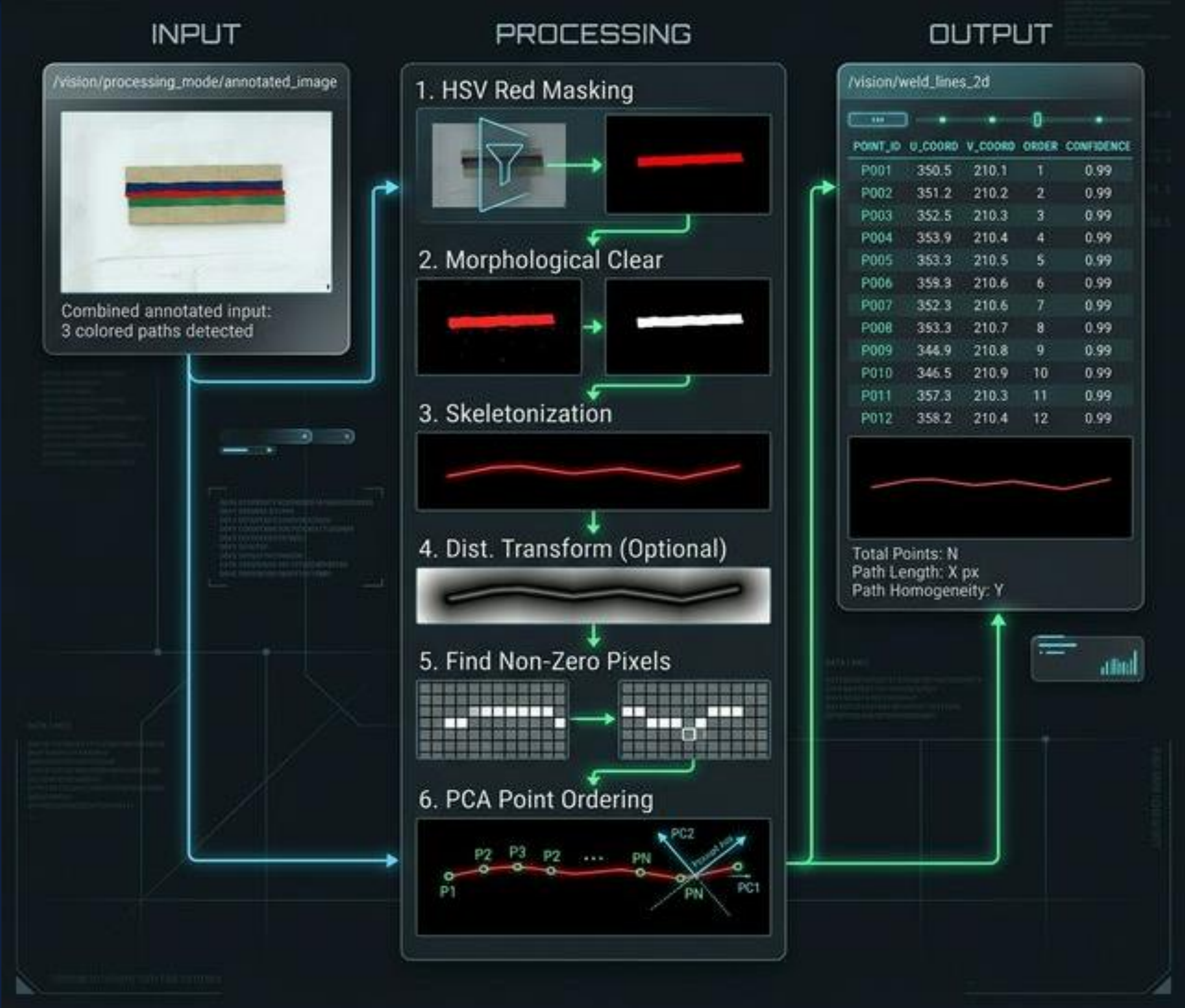


Manual Mode

Stage 4: Red Line Optimization

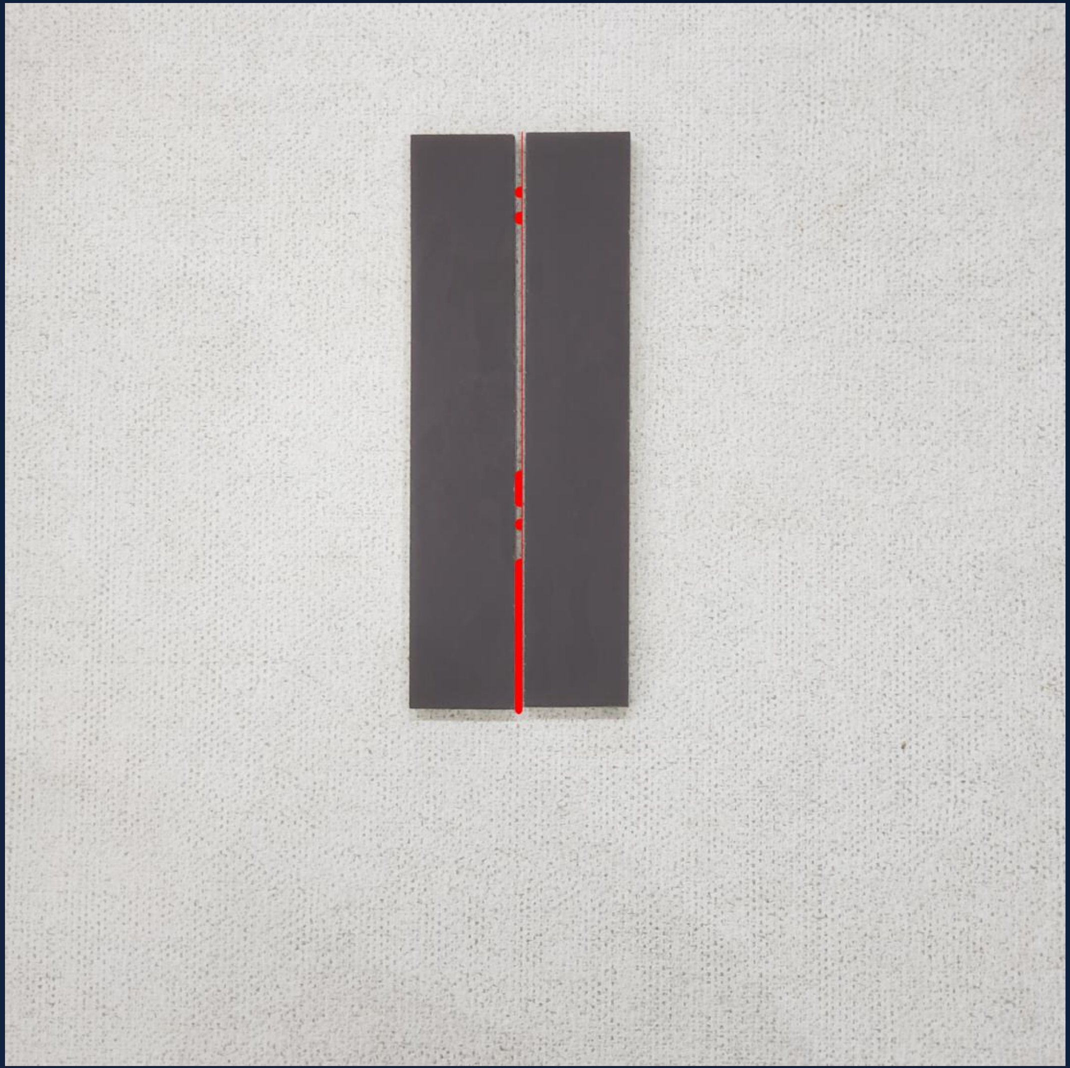
Detects the red weld marker line(s) painted on the workpiece surface (or generated by manual mode) within the annotated image from Stage 3.

It applies a multi-step computer vision pipeline and outputs a structured "WeldLineArray" message with ordered pixel-space points representing the weld path.

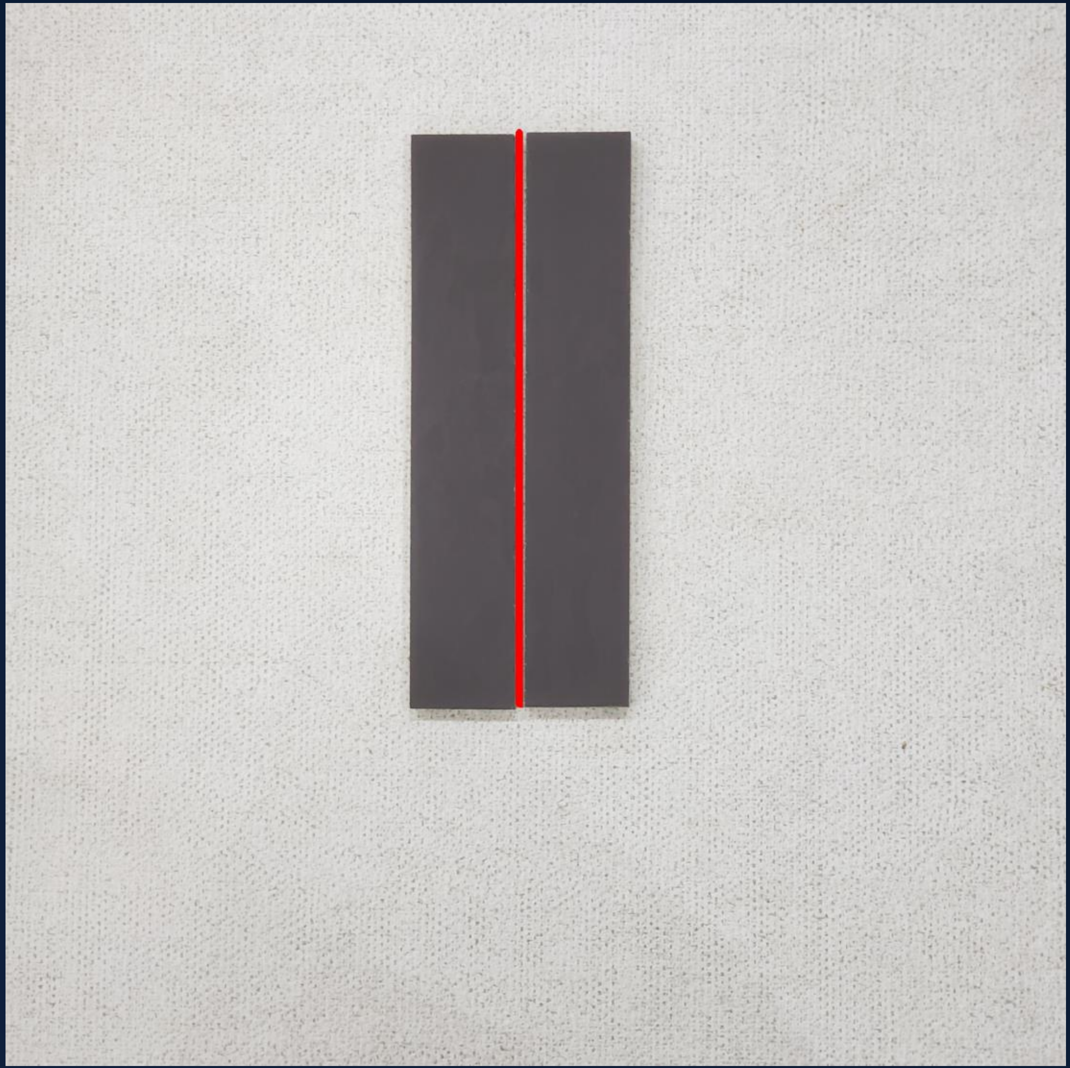


Stage 4: Red Line Optimization (Example)

Input (from Stage 3)



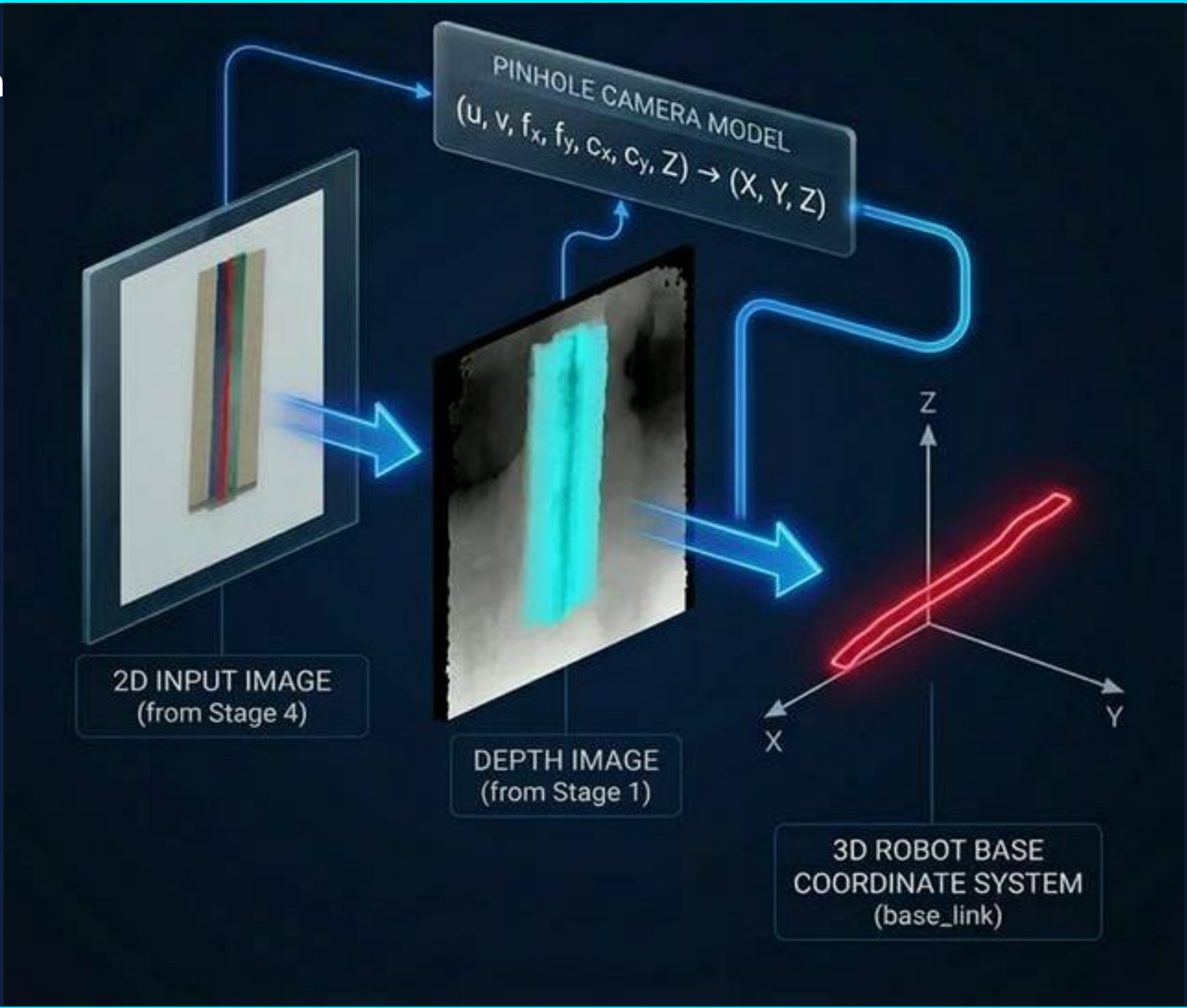
Output (to Stage 5)



Stage 5: 2D-to-3D Reconstruction

The critical bridge between the 2D vision world and 3D robot space.

It takes the ordered pixel-space weld line points from Stage 4 and back-projects each pixel into a 3D Cartesian coordinate using the depth camera's intrinsic matrix and the robot's TF2 transform tree.



Stage 6: Path Generation

Converts the raw 3D point cloud of the weld seam into a smooth, kinematically feasible welding trajectory (nav_msgs/Path).
It applies PCA ordering, duplicate removal, cubic B-spline fitting, arc-length resampling, and 6-DOF orientation assignment to produce a trajectory ready for MoveIt2 execution.